

Student Name: _____ **PC #:** _____

ECCS 4441 – Ethical Hacking and Penetration Testing

PROJECT 2: ADVANCED NETWORK RECONNAISSANCE AND STEALTH SCANNING WITH NMAP

PROJECT OVERVIEW

In this project, you will act as a penetration tester performing a controlled network reconnaissance assessment using Nmap. The goal is to develop deep technical proficiency with Nmap scanning techniques, understand how different operating systems respond to scans, and evaluate how scan activity can be detected or evaded.

LEARNING OBJECTIVES

By completing this project, you will be able to:

- Apply Nmap to discover hosts, open ports, services, and operating systems
- Use and explain a wide variety of Nmap scan switches
- Compare scan behavior across multiple operating systems
- Perform stealth scans and analyze their effectiveness
- Identify how targets detect Nmap scanning activity
- Interpret scan results and network traffic
- Produce a professional penetration testing report

LAB ENVIRONMENT

Virtual Machines (VMware Workstation Pro)

Virtual Machine	Role
Kali Linux	Attacker / Scanner
Windows 10	Target System
Windows 7	Legacy Target System
Metasploitable 2	Vulnerable Target

Network Configuration

- All VMs must be on the NAT
- Do not scan any system outside this lab

PROJECT REQUIREMENTS

You must complete all phases of the project and submit the required deliverables. Screenshots must clearly show commands and results.

PHASE 1: NETWORK DISCOVERY

Objective

Identify live hosts on the network using multiple discovery techniques.

Tasks

- Determine the IP address range of your lab network (i.e. your computer virtual environment)
- Perform host discovery using different Nmap options
- Do not check the IPs of all three target machines. Discover them using nmap.
- Turn off the firewalls of both Windows 7 and 10 VMs.
- First, Power ON one VM (Windows 7), then discover using Kali. Keep it Powered ON.
- Second, Power ON Windows 10, and discover. Keep it Powered ON
- Third, Power ON Metasploitable 2 and discover. Keep it Powered ON.

Example Commands

```
nmap -sn <network_range>  
nmap -Pn <target_ip>
```

Deliverables

- List of discovered hosts
- Explanation of each discovery method
- Comparison of results across targets

PHASE 2: PORT SCANNING AND SERVICE ENUMERATION

Objective

Identify open ports, services, and operating systems.

Required Scan Types

Perform the following scans against each target VM:

- TCP Connect Scan (-sT)
- SYN Scan (-sS)

- UDP Scan (-sU)
- Service Version Detection (-sV)
- OS Detection (-O)
- Aggressive Scan (-A)

Example Commands

```
nmap -sS -p- <target_ip>
nmap -sV -O <target_ip>
nmap -A <target_ip>
```

Deliverables

- Table of open ports and services per target
- OS detection results and accuracy discussion
- Explanation of differences between targets

PHASE 3: STEALTH SCANNING TECHNIQUES

Objective

Conduct scans designed to reduce detection by target systems.

Required Stealth Scans

- FIN Scan (-sF)
- NULL Scan (-sN)
- Xmas Scan (-sX)
- Fragmented Packets (-f)
- Decoy Scanning (-D)
- Slow Timing Templates (-T0 to -T2)

Example Commands

```
nmap -sN <target_ip>
nmap -sF -f <target_ip>
nmap -sS -T1 -D RND:5 <target_ip>
```

Deliverables

- Comparison of stealth vs standard scan results
- Identification of scans that were blocked or allowed
- Explanation of why certain scans are stealthier

PHASE 4: DETECTION AND DEFENSE ANALYSIS

Objective

Analyze how target systems detect Nmap scans.

Tasks

- Enable firewall logging on Windows 10 and Windows 7
- Capture network traffic using Wireshark on a target VM
- Identify scan signatures such as:
 - Half-open connections
 - Unusual TCP flag combinations
 - Rapid port probing

Deliverables

- Screenshots of firewall alerts or logs
- Wireshark packet capture screenshots
- Explanation of which scans were most detectable

PHASE 5: VULNERABILITY MAPPING (METASPLOITABLE 2)

Objective

Relate Nmap findings to known vulnerabilities.

Tasks

- Run vulnerability detection scripts
- Identify vulnerable services
- Explain how Nmap supports exploitation planning

Example Command

```
nmap --script vuln <metasploitable_ip>
```

Deliverables

- List of identified vulnerabilities
- Relevant CVE references
- High-level risk discussion (no exploitation required)

PHASE 6: FINAL REPORT

Report Requirements

Your report must be organized, clear, and the results should be supported by evidence (screenshots).

Required Sections

1. Executive Summary (half to one page)
2. Scope and Methodology
3. Network Discovery Results
4. Scan Results by Target System
5. Stealth and Detection Analysis
6. Risk Assessment
7. Mitigation Recommendations
8. Nmap Commands Used within each Scan Results.

SUBMISSION REQUIREMENTS

- Final report (PDF format), upload to Canvas.
- Deadline: Tentatively February 19th, 2026.
- Screenshots included in report